

Immunogenicity and safety of concomitant vaccines given with 20-valent pneumococcal conjugate vaccine in healthy infants

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ABSTRACT

Background: Pneumococcal conjugate vaccines (PCVs) are generally administered in infant schedules with other vaccines. Here, we describe the safety and immunogenicity of routinely recommended pediatric vaccines given with 20-valent-PCV (PCV20).

Methods: Two phase 3, double-blind, randomized trials evaluated the safety, tolerability, and immunogenicity of PCV20 in infants with concomitant pediatric vaccines. The studies examined a 4-dose series (Study B7471011/NCT04382326; $N = 1997$) and a 3-dose series (Study B7471012/NCT04546425; $N = 1207$). Concomitant vaccinations included a combination vaccine (diphtheria, tetanus, acellular pertussis [DTaP] with hepatitis B and poliovirus antigens [DTaP-HBV-IPV] and *Haemophilus influenzae* type b vaccine [Hib]) in Study B7471011 or combined vaccine (DTaP, hepatitis B, poliovirus, and Hib antigens [DTaP-HBV-IPV/Hib]) in Study B7471012. Measles, mumps, rubella (MMR) and varicella vaccines were given concomitantly with the toddler dose in both studies. Immunogenicity objectives were to demonstrate noninferiority of immune responses to concomitant vaccine antigens for concomitant vaccines given with PCV20 to those given with 13-valent PCV (PCV13). Safety endpoints included systemic events and adverse events (AEs).

Results: Noninferiority was met for the primary objectives of percentage of participants with prespecified antibody levels to the DTaP-HBV-IPV and Hib 1 month after the third infant dose (Study B7471011) and 1 month after the toddler dose of DTaP-HBV-IPV/Hib (Study B7471012) in the PCV20 group compared with the PCV13 group. Noninferiority was also met for the geometric mean antibody levels to MMR and varicella vaccines 1 month after the toddler dose in both studies. Systemic event frequencies were similar in the PCV20 and PCV13 groups in both studies, with severity mostly reported as mild or moderate. Frequencies of reported AEs were similar between the PCV20 and PCV13 groups.

Conclusions: Immune responses to routine pediatric vaccines given with PCV20 were noninferior to those when given with PCV13. PCV20 may be safely administered with other routine pediatric vaccines. (NCT04382326; NCT04546425).

Abbreviations: AE, adverse event; DTaP, diphtheria, tetanus, acellular pertussis; DTaP-HBV-IPV, vaccine containing DTaP with hepatitis B and poliovirus antigens; DTaP-HBV-IPV/Hib, vaccine containing DTaP, hepatitis B, poliovirus, and Hib antigens; FHA, filamentous hemagglutinin; GM, geometric mean; GMR, geometric mean ratio; HBsAg, hepatitis B surface antigen; Hib, *Haemophilus influenzae* type b vaccine; IPD, invasive pneumococcal disease; MMR, measles, mumps, rubella; PCV, pneumococcal conjugate vaccine; PCV13, 13-valent pneumococcal conjugate vaccine; PCV20, 20-valent pneumococcal conjugate vaccine; PRN, pertactin; PT, detoxified pertussis toxin; SAE, serious adverse event.

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1. Introduction

Pneumococcal disease is a significant cause of morbidity and mortality in children younger than 5 years and can lead to noninvasive conditions, such as acute otitis media, pneumonia, or life-threatening invasive pneumococcal disease (IPD), including bacteremia and bacterial meningitis [1–3]. Widespread implementation and uptake of pneumococcal conjugate vaccines (PCVs) through national infant immunization programs have substantially decreased the burden of pneumococcal disease worldwide [4,5].

The first widely used PCV was 7-valent PCV (PCV7; serotypes 4, 6B, 9V, 14, 18C, 19F, 23F), which was licensed in 2000 and subsequently replaced with the 13-valent PCV (PCV13; 6 additional serotypes, 1, 3, 5, 6A, 7F, 19A) within infant immunization programs in 2010 [6,7]. Although uptake of PCVs averted an estimated 175 million global cases of pneumococcal disease and more than 620,000 associated deaths in children younger than 5 years, a substantial disease burden due to non-PCV13 serotypes has remained [4,8–11].

To help address the need for expanded protection against additional serotypes, a 20-valent PCV (PCV20) containing PCV13 components and conjugates for 7 additional serotypes (8, 10A, 11A, 12F, 15B, 22F, 33F) was developed [12–19]. These additional serotypes were selected based on their global prevalence as a cause of pediatric pneumococcal disease and their association with increased disease severity, invasive potential, and antibiotic resistance [8,20–24]. PCV20 has been licensed in several countries for the prevention of pneumococcal pneumonia and IPD among adults ≥ 18 years of age, and for pediatric use from 6 weeks of age [25–30].

Two key phase 3 infant immunogenicity and safety trials demonstrated that PCV20 elicited immune responses expected to protect infants against all vaccine serotypes and had a similar safety and tolerability profile to that of PCV13 [17,18]. Here, we expand on the findings from these studies on the immunogenicity and safety of routinely recommended pediatric vaccines administered concomitantly with PCV20.

2. Methods

2.1. Study design and participants

The study designs and ethical conduct of 2 phase 3, double-blind, randomized trials that evaluated the safety, tolerability, and immunogenicity of PCV20 in infants when administered concomitantly with routine childhood vaccines have been reported previously (**Supplementary Fig. 1**) [17,18]. Briefly, Study B7471011 (NCT04382326) evaluated 4 doses (3 infant doses plus a toddler dose) of PCV20, compared with PCV13, and enrolled 1997 participants from 92 sites in the United States and Puerto Rico from May 2020 to September 2022 [17]. Study B7471012 (NCT04546425) evaluated 3 doses (2 infant doses plus a toddler dose) of PCV20, compared with PCV13, and enrolled 1207 participants from 59 sites in Europe and Australia (primary analysis population) from September 2020 to April 2022 [18]. Participants were healthy infants born at >36 weeks of estimated gestational age who were approximately 2 months of age at the time of consent. Both studies excluded infants with previous receipt of the following vaccines: pneumococcal, diphtheria, tetanus, pertussis, poliomyelitis, or *Haemophilus influenzae* type b (Hib). Additional eligibility criteria and methodology have been previously reported [17,18].

Participants were randomized in a 1:1 ratio to receive PCV20 or PCV13 at approximately 2, 4, 6, and 12–15 months of age (Study B7471011) [17], and at approximately 2, 4, and 11–12 months of age (Study B7471012) [18]. Participants in Study B7471011 received diphtheria, tetanus, and acellular pertussis vaccine (DTaP) in combination with poliovirus and hepatitis B antigens (PEDIARIX; GlaxoSmithKline) and Hib (HIBERIX; GlaxoSmithKline) concomitantly with PCV20 or PCV13 at Doses 1, 2, and 3; measles, mumps, and rubella

(MMR; M-M-R_{II}; Merck) and varicella (VARIVAX; Merck) vaccines were administered with Dose 4 [17]. Participants in Study B7471012 received a combined vaccine containing DTaP, hepatitis B, poliovirus, and Hib antigens (Infanrix-hexa; GlaxoSmithKline) concomitantly with all 3 doses of PCV20 or PCV13. Measles, mumps, and rubella (MMRVaxPro, Merck Sharpe & Dohme), and varicella vaccines (Varilrix, GlaxoSmithKline) were administered with the toddler dose (Dose 3) except where local guidelines made this impractical. In Study B7471012, certain sites elected not to administer the MMR or varicella vaccines based on local recommendations; however, most participants received both vaccines [18]. Rotavirus and influenza vaccines were permitted to be given at any time (including with PCV20 or PCV13) in both studies, in accordance with local guidelines [17,18].

2.2. Objectives and endpoints

2.2.1. Immunogenicity

The primary concomitant immunogenicity objective for both studies was to demonstrate that immune responses elicited by specified concomitant vaccine antigens given with PCV20 were noninferior to immune responses elicited by concomitant vaccine antigens given with PCV13. In both studies, specified antigens were diphtheria toxoid, tetanus toxoid, pertussis antigens (filamentous hemagglutinin, pertactin, pertussis toxoid), hepatitis B surface antigen, poliovirus strains, and Hib. Responses were assessed at 1 month after Dose 3 (third infant dose) for DTaP, hepatitis B, poliovirus, and Hib antigens in Study B7471011 (4-dose series), and 1 month after Dose 3 (toddler dose) in Study B7471012 (3-dose series). Immune responses to MMR and varicella antigens were assessed after the toddler dose in both studies.

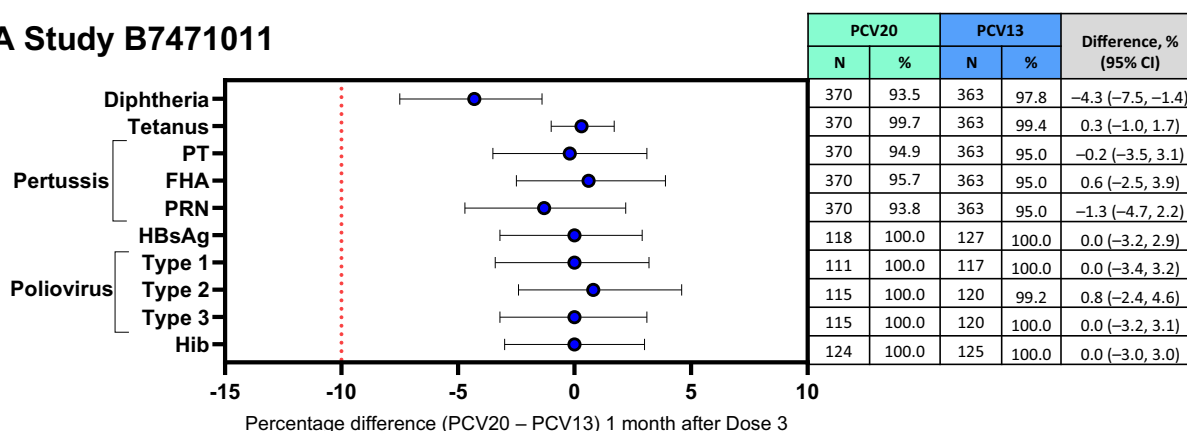
Participants were randomly selected for measurement and assessment of concomitant vaccine responses and included if they remained in the evaluable population, with the sample sizes of the subsets predetermined (varying by antigen) to ensure adequate power to show noninferiority for the specified analyses. Blood samples were collected at defined intervals (**Supplementary Fig. 1**), and antibody responses to concomitant vaccine antigens were measured using a validated multiplex Luminex immunoassay, chemiluminescent immunoassay, antibody microneutralization test, or immunoglobulin G enzyme-linked immunosorbent assay as appropriate for the concomitant vaccine antigen.

Noninferiority was demonstrated based on comparing results of the concomitant immunogenicity endpoints: 1) percentages of participants with prespecified antibody levels to DTaP, hepatitis B, poliovirus, and Hib antigens for each study at the time points noted above with prespecified levels of ≥ 0.1 IU/mL for diphtheria and tetanus, pertussis (PT, FHA, PRN) at or above the observed antipertussis concentration achieved by 95 % of participants in the PCV13 group for pertussis, ≥ 10 mIU/mL for hepatitis B surface antigen, $\geq 1:8$ for poliovirus (types 1, 2, and 3), and ≥ 0.15 μ g/mL for Hib, using a 10 % noninferiority criterion; and 2) geometric mean antibody levels for all concomitant vaccine antigens to MMR and varicella antigens 1 month after the toddler dose using a 2-fold noninferiority criterion.

2.2.2. Safety

In the 2 studies, safety was evaluated; safety endpoints included prespecified local reactions and systemic events within 7 days, adverse events (AEs), and serious AEs (SAEs). Local reactions were assessed only for the injection site of PCV20 and PCV13 administration and not for concomitant vaccines. In Study B7471011, AEs were assessed from Dose 1 through 1 month after Dose 3 and from Dose 4 through 1 month after Dose 4; SAEs were collected through 6 months after Dose 4. In Study B7471012, AEs were assessed from Dose 1 through 1 month after Dose 2 and from Dose 3 through 1 month after Dose 3; SAEs were collected through 1 month after Dose 3. Post hoc exploratory descriptive safety subgroup analyses presented here included participants who did and did not receive the influenza vaccine in Study B7471011 or rotavirus vaccine in Study B7471012 at study vaccine visits as part of age-appropriate

A Study B7471011



B Study B7471012

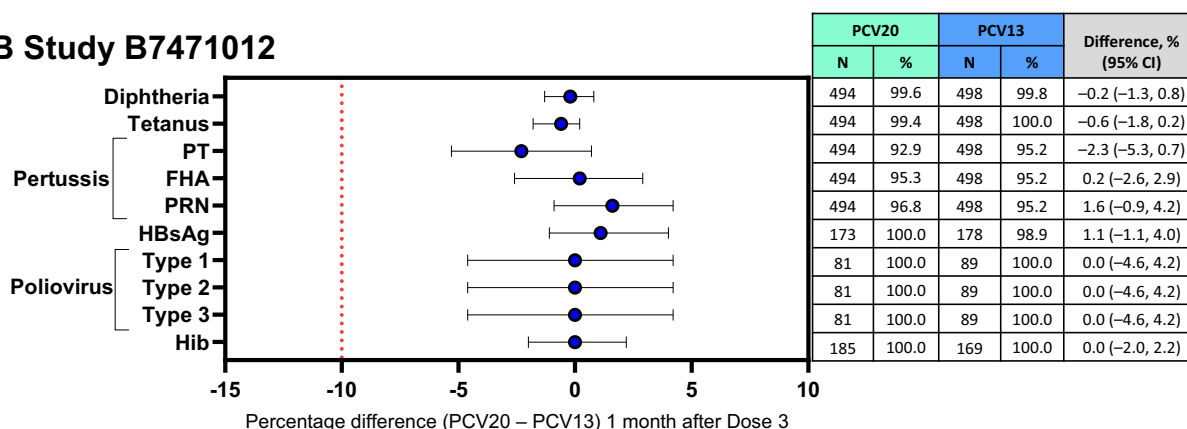


Fig. 1. Differences (2-Sided 95 % CIs) in Percentages of Participants With Prespecified Antibody Levels for DTaP, Poliovirus, Hepatitis B, and Hib Antigens 1 Month After (A) Dose 3 (Third Infant Dose) in Study B7471011 and (B) Dose 3 (Toddler Dose) in Study B7471012. Immune responses to each concomitant vaccine antigen were determined on sera from randomly selected subsets of participants in both vaccine groups in each study. The dotted line (at the percentage difference of -10 %) represents the 10 % noninferiority criterion. N is the number of participants with valid assay results for the specified concomitant vaccine antigen among those in the evaluable immunogenicity population who received the corresponding concomitant vaccine. These values are the denominators for the percentage calculations. FHA, filamentous hemagglutinin; HBsAg, hepatitis B surface antigen; Hib, *Haemophilus influenzae* type b; N, number of participants with valid assay results for the specified antigen; PCV13, 13-valent pneumococcal conjugate vaccine; PCV20, 20-valent pneumococcal conjugate vaccine; PRN, pertactin; PT, pertussis toxoid.

routine care.

2.3. Statistical analysis

Noninferiority of immune responses induced by a concomitant vaccine antigen given with PCV20 to those with PCV13 for prespecified immune responses was declared if the lower bound of the 2-sided 95 % CI for the difference (PCV20 group – PCV13 group) in the percentages of participants with the prespecified antibody level exceeded -10 % (10 % noninferiority criterion). Noninferiority of the PCV20 group to the PCV13 group for the immune response to each of MMR and varicella virus antigen was declared if the lower bound of the 2-sided 95 % CI of the geometric mean ratio (GMR; PCV20 group to PCV13 group) was >0.5 (2-fold noninferiority criterion).

The evaluable immunogenicity population for each analysis included participants who were within the protocol-defined age windows for the first and last dose, received 3 or 4 doses (Study B7471011) or 3 doses (Study B7471012) of pneumococcal vaccine as randomized in addition to the concomitant vaccine, had ≥1 valid immunogenicity result from blood collected within a prespecified window 1 month after Dose 3 or 4 (Study B7471011) or Dose 3 (Study B7471012), and had no other major protocol deviations determined by the study clinician.

Overall safety data were analyzed by descriptive summary statistics. The safety populations included all participants who received any study

vaccination. The respective exploratory safety analyses were also conducted by subgroups of participants based on whether they had received influenza vaccine or not at Dose 3 or Dose 4 (Study B7471011) for the influenza vaccine safety assessment or whether they received rotavirus vaccine with both Doses 1 and 2 (Study B7471012) for the rotavirus vaccine safety assessment.

3. Results

3.1. Participants

In Study B7471011, 1997 participants were randomized to receive 4 doses of study vaccine, with 1991 receiving either PCV20 ($n = 1001$) or PCV13 ($n = 990$); of these, 1860 (93.1 %) and 1697 (85.0 %) participants completed Dose 3 and Dose 4, respectively. In Study B7471012, 1207 participants were randomized to receive 3 doses of vaccine and included in this concomitant vaccine analysis, with 1204 receiving either PCV20 ($n = 601$) or PCV13 ($n = 603$); of these, 1182 (97.9 %) received all 3 doses. As previously reported for both studies, the PCV20 and PCV13 groups had similar overall participant demographic characteristics [17,18].

Concomitant vaccine administration in participants for both studies is shown by study vaccine visit in **Table S1**; the majority of participants in both studies received the prespecified concomitant vaccines per

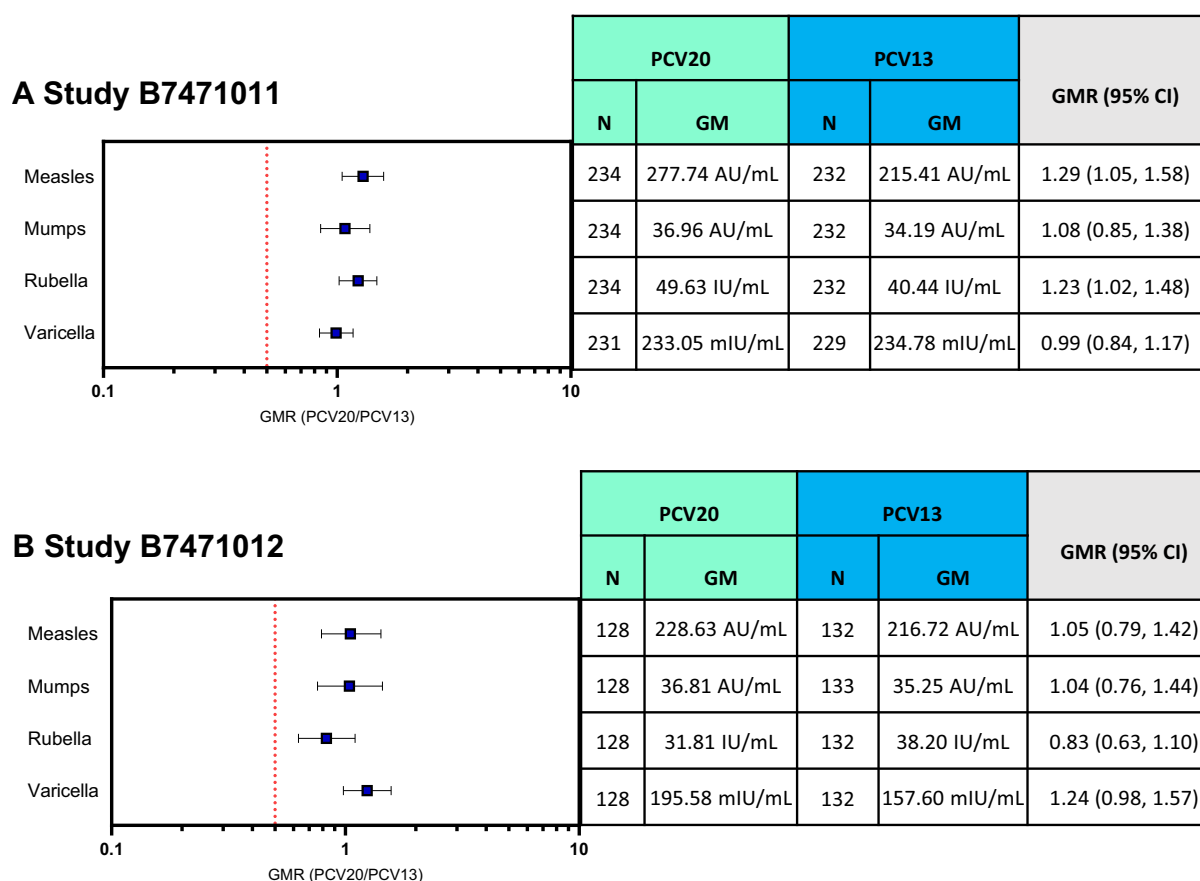


Fig. 2. GMRs (2-Sided 95 % CIs) for Measles, Mumps, Rubella, and Varicella Antigens 1 Month After the Toddler Dose: Dose 4 in Study B7471011 (A) and Dose 3 in Study B7471012 (B). Immune responses to each concomitant vaccine antigen were determined on sera from randomly selected subsets of participants in both vaccine groups in each study. The dotted line (at the GMR of 0.5) represents the 2-fold noninferiority criterion. Geometric means, GMRs (PCV20/PCV13), and 2-sided CIs were calculated by exponentiating the mean/mean difference logarithm of the concentrations and the corresponding CIs (based on the Student *t* distribution). GM, geometric mean; GMR, geometric mean ratio; N, number of participants with valid assay results for the specified antigen; PCV13, 13-valent pneumococcal conjugate vaccine; PCV20, 20-valent pneumococcal conjugate vaccine.

protocol. Additionally, in Study B7471011, 10 % to 12 % of participants received influenza vaccine with Dose 3 (PCV20: 11.7 %; PCV13: 12.3 %) or Dose 4 (PCV20: 10.1 %; PCV13: 11.6 %); and the majority of participants received rotavirus vaccine with Dose 1 (PCV20, 87.3 %; PCV13, 87.4 %), Dose 2 (PCV20, 84.9 %; PCV13, 84.0 %), and Dose 3 (PCV20, 65.8 %; PCV13, 64.9 %). In Study B7471012, influenza vaccine was coadministered with Dose 3 in <1 % of participants overall; over half of the participants received rotavirus vaccine with Dose 1 (PCV20: 56.9 %; PCV13: 56.4 %) or Dose 2 (PCV20: 55.1 %; PCV13: 57.7 %). Approximately 47 % of participants received rotavirus vaccine at Dose 1 or Dose 2 in both groups in Study B7471012.

3.2. Immunogenicity

Noninferiority for concomitant immune responses of the PCV20 group to the PCV13 group was demonstrated for the percentage of participants with prespecified antibody levels to all antigens in the DTaP-HBV-IPV and Hib 1 month after the third infant dose (Dose 3) in Study B7471011 (Fig. 1A) with the noninferiority criterion (lower bound of 2-sided 95 % CI of percentage difference greater than −10 %) met for all comparisons. Similarly, noninferiority for concomitant immune responses of the PCV20 group to the PCV13 group after the toddler dose (Dose 3) was demonstrated when given with DTaP-HBV-IPV/Hib in Study B7471012 (Fig. 1B).

For both studies, geometric mean antibody levels were similar 1 month after Dose 3 in the PCV20 and PCV13 groups for all antigens in DTaP-HBV-IPV and Hib (Study B7471011) and DTaP-HBV-IPV/Hib

(Study B7471012) (Table S2).

Noninferiority for concomitant immune responses of the PCV20 group to the PCV13 group was demonstrated for MMR and varicella vaccine antigens (2-sided 95 % CIs lower bounds of GMRs >0.5) after the toddler dose in Study B7471011 (Fig. 2A) and Study B7471012 (Fig. 2B).

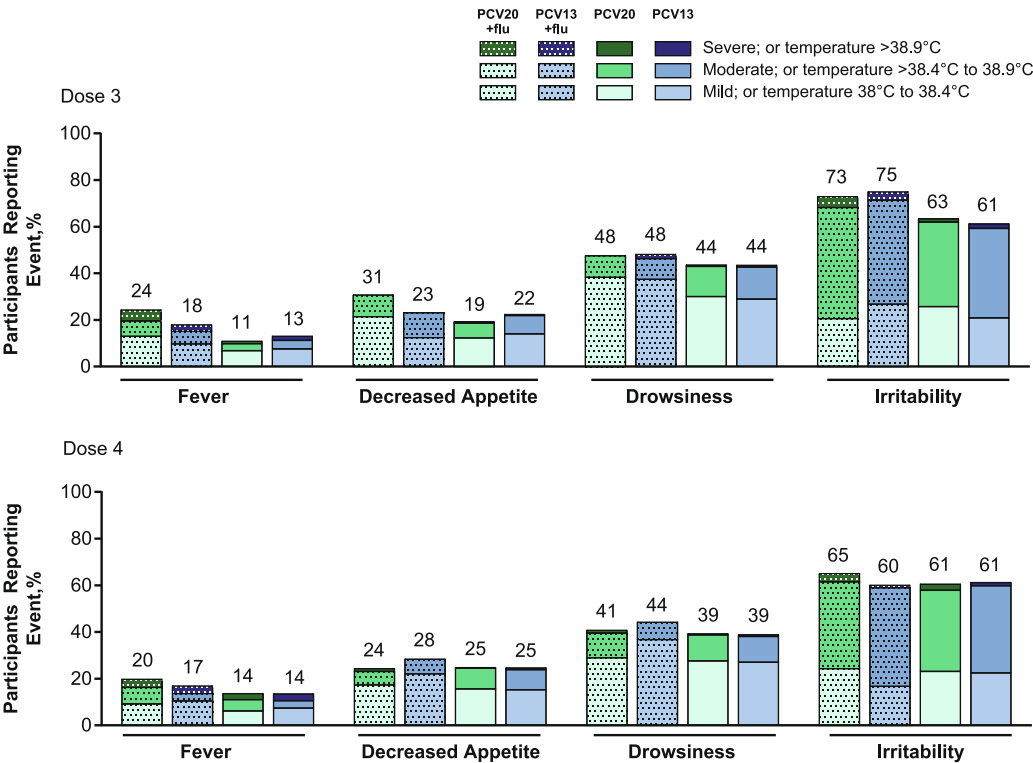
3.3. Safety

The overall safety results for PCV20 and PCV13 given in the setting of the concomitant pediatric vaccines specified for Study B7471011 and Study B7471012 have been fully reported previously [17,18]. PCV20 had a similar tolerability and safety profile as PCV13 in both studies, with mild to moderate reactogenicity and AEs generally consistent with illnesses and medical conditions expected in infant and toddler populations [17,18]. There were no SAEs considered by the investigator to be related to PCV20 in Study B7471011 [17]. In Study B7471012, a previously reported SAE of inflammation 7 days after Dose 1 of PCV20 was considered by the investigator to be potentially related to PCV20 or DTaP-HBV-IPV/Hib combination vaccine [18].

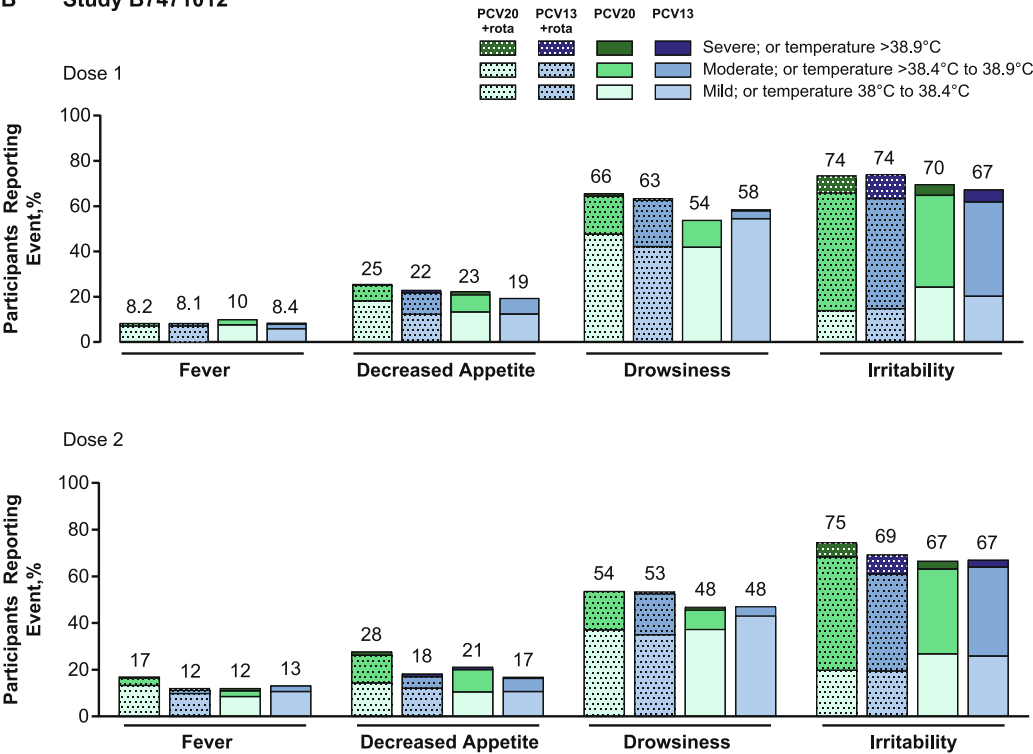
3.3.1. Post hoc subgroup analyses

The number of participants who received either influenza vaccine in Study B7471011 or rotavirus in Study B7471012, permitted at age-appropriate study visits, was reasonable in each stated subset to allow an exploratory analysis of the potential impact of those vaccines on the safety profile.

A Study B7471011



B Study B7471012



(caption on next page)

Fig. 3. Percentages of Participants Reporting Prespecified Systemic Events Within 7 Days by Receipt of (A) Influenza Vaccine in Study B7471011 and (B) Rotavirus Vaccine in Study B7471012. Fever parameters are defined within the figure. Decreased appetite defined as mild, decreased interest in eating; moderate, decreased oral intake; and severe, refusal to feed. Drowsiness defined as mild, increased or prolonged sleeping bouts; moderate, slightly subdued interfering with daily activity; and severe, disabling not interested in usual daily activity. Irritability defined as mild, easily consolable; moderate, requiring increased attention; and severe, inconsolable, crying cannot be comforted. Values above the bars show the percentages of participants reporting any event. The number of participants per subgroup for each dose shown was as follows: PCV20 + flu, $n = 86$ –107; PCV13 + flu, $n = 95$ –112; PCV20, $n = 740$ –807; PCV13, $n = 720$ –789 (Panel A); PCV20 + rota, $n = 278$ –279; PCV13 + rota, $n = 282$ –284; PCV20, $n = 209$ –210; PCV13, $n = 197$ –202 (Panel B). PCV13, 13-valent pneumococcal conjugate vaccine; PCV20, 20-valent pneumococcal conjugate vaccine; flu, influenza vaccine; rota; rotavirus vaccine.

Table 1

Summary of System Organ Classes in Which $\geq 2\%$ of Recipients in Any Group Reported AEs: by Receipt of Influenza Vaccine in Study B7471011 and by Receipt of Rotavirus Vaccine in Study B7471012.

Study B7471011	With influenza vaccine		Without influenza vaccine	
	PCV20	PCV13	PCV20	PCV13
System organ class				
Dose 3 to 1 month after Dose 3, N	109	113	825	811
Any event, n (%)	19 (17.4)	15 (13.3)	107 (13.0)	120 (14.8)
Gastrointestinal disorders	3 (2.8)	2 (1.8)	11 (1.3)	15 (1.8)
General disorders and administration site conditions	3 (2.8)	1 (0.9)	4 (0.5)	3 (0.4)
Infections and infestations	10 (9.2)	9 (8.0)	72 (8.7)	83 (10.2)
Skin and subcutaneous tissue disorders	3 (2.8)	2 (1.8)	16 (1.9)	11 (1.4)
Dose 4 to 1 month after Dose 4, N	86	98	767	743
Any event, n (%)	15 (17.4)	21 (21.4)	114 (14.9)	105 (14.1)
Infections and infestations	13 (15.1)	15 (15.3)	93 (12.1)	74 (10.0)
Investigations	1 (1.2)	3 (3.1)	5 (0.7)	3 (0.4)
Respiratory, thoracic, and mediastinal disorders	2 (2.3)	4 (4.1)	14 (1.8)	10 (1.3)
Skin and subcutaneous tissue disorders	1 (1.2)	2 (2.0)	10 (1.3)	15 (2.0)
Study B7471012				
System organ class				
Dose 1 to 1 month after Dose 2, N	279	284	211	202
Any event, n (%)	55 (19.7)	48 (16.9)	15 (7.1)	25 (12.4)
Gastrointestinal disorders	4 (1.4)	10 (3.5)	0 (0.0)	1 (0.5)
General disorders and administration site conditions	4 (1.4)	10 (3.5)	0 (0.0)	0 (0.0)
Infections and infestations	35 (12.5)	25 (8.8)	9 (4.3)	16 (7.9)
Respiratory, thoracic, and mediastinal disorders	6 (2.2)	2 (0.7)	1 (0.5)	1 (0.5)
Skin and subcutaneous tissue disorders	10 (3.6)	9 (3.2)	1 (0.5)	3 (1.5)

AE, adverse event; N, number of participants in the specified group at each dose; n, number of participants reporting ≥ 1 occurrence of the specified event; PCV13, 13-valent pneumococcal conjugate vaccine; PCV20, 20-valent pneumococcal conjugate vaccine; SAE, serious adverse event.

The percentages of participants with systemic events were generally similar in the PCV20 and PCV13 groups regardless of whether the study vaccines were administered with or without influenza vaccine in Study B7471011 (Fig. 3A). Fever $\geq 38.0^\circ\text{C}$ was reported in 16.8 %–24.3 % across groups of participants who also received influenza vaccine with Doses 3 or 4 and 11.0 %–13.9 % in those who did not. Similar percentages of systemic events were observed among participants in Study B7471012 regardless of whether the study vaccines were administered with or without rotavirus vaccine (Fig. 3B). Fever $\geq 38.0^\circ\text{C}$ was reported in 8.1 %–16.9 % in participants who also received rotavirus vaccine with Doses 1 and 2 and 8.4 %–13.2 % in those not receiving rotavirus vaccine with both doses.

In Study B7471011, AEs were reported at generally similar rates in the PCV20 and PCV13 groups regardless of whether participants received influenza vaccine (Table 1). There were no signals for an increase in febrile seizures among participants who received concomitant influenza vaccine administration compared with those who did not receive influenza vaccine in either the PCV13 or PCV20 groups.

In Study B7471012, from Dose 1 to 1 month after Dose 2, AEs were reported at generally similar rates in PCV20 and PCV13 groups, with or without concomitant rotavirus vaccine (Table 1). In those who received concomitant rotavirus vaccine, AEs were reported in 19.7 % and 16.9 % of participants in the PCV20 and PCV13 groups, respectively; whereas AEs were reported in 7.1 % of PCV20 recipients and 12.4 % of PCV13 recipients who did not receive rotavirus vaccine. The higher frequency of AEs in the PCV20 subgroup with rotavirus vaccine was driven primarily by unrelated AEs in the system organ class of infections and infestations (12.5 % and 4.3 %, respectively) and included conjunctivitis, upper respiratory tract infection, and nasopharyngitis, which are likely to be incidental findings.

4. Discussion

Guidance on routine pediatric immunizations advises administration of multiple vaccines concomitantly to enhance vaccine uptake and compliance, as well as reduce burden on both healthcare systems and individuals [31]. Incorporation of PCV20 into infant immunization programs has the potential to expand protection for children against PCV serotypes by protecting against additional serotypes while also maintaining established protection against disease from PCV13 serotypes. Importantly, the clinical development program for PCV20 included assessment of safety and immunogenicity of routine concomitant pediatric vaccines.

In 2 phase 3 trials in infants, noninferiority criteria were met for the percentages of participants with prespecified antibody levels for all antigens within DTaP-HBV-IPV and Hib (Study B7471011) or DTaP-HBV-IPV/Hib (Study B7471012) in the PCV20 group compared with the PCV13 group. In both studies, geometric mean antibody levels to MMR and varicella vaccines were also noninferior for PCV20 to PCV13 groups.

Previously reported safety results from these studies have shown that PCV20 was well tolerated with a safety profile similar to PCV13 in the setting of use with routine pediatric vaccines [17,18]. Here, we provide exploratory subgroup analyses of systemic events and AEs with and without concomitant influenza and rotavirus vaccines. The safety profile of PCV20 was generally similar when given with or without influenza and rotavirus vaccines.

Findings from these trials can potentially reflect real-world use of PCV20 and incorporation into routine vaccination schedules for infants. This analysis does not differentiate between systemic events and AEs arising from either PCV20 or PCV13 compared with concomitant vaccines, potentially confounding severity responses across vaccines. However, it would not be feasible or ethical to delay vaccination or include a control group without a PCV. Moreover, the study design reflects real-world scenarios wherein infants receive multiple pediatric vaccinations simultaneously, demonstrating that most events were mild to moderate when PCV20 was administered with concomitant pediatric vaccines. Additionally, subgroup safety data are described for participants receiving concomitant influenza and rotavirus vaccines. These

were assessed retrospectively in nonrandomized populations with unequal sample sizes and were exploratory in nature; conclusions arising from these exploratory data should be interpreted with caution. Finally, it is worth noting that the 2 randomized trials were not powered to detect and attribute rarer adverse events. Ongoing surveillance and real-world studies will continue to monitor safety.

In conclusion, immune responses to routine concomitant pediatric vaccines given with PCV20 in infants and children in both studies were noninferior to responses when given with PCV13. Additionally, these results support the safe coadministration of PCV20 with common pediatric vaccines in this analysis and inclusion of PCV20 in routine infant immunization schedules.

CRediT authorship contribution statement

Shelly Senders: Investigation. **Piotr Korbal:** Investigation. **Mary Kline:** Formal analysis, Methodology, Supervision. **Noor Tamimi:** Formal analysis, Investigation, Methodology, Supervision. **Allison Thompson:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision. **Jelena Drozd:** Investigation. **Mark W. Cutler:** Investigation, Methodology. **Peter C. Giardina:** Investigation, Methodology. **James Trammel:** Conceptualization, Formal analysis, Methodology. **Lanyu Lei:** Conceptualization, Formal analysis, Methodology. **Yahong Peng:** Conceptualization, Methodology, Supervision. **Wendy Watson:** Conceptualization, Methodology, Supervision. **Kathleen McElwee:** Formal analysis.

Patient consent

Parental informed consent was obtained after the nature and possible consequences of the study had been fully explained to the parents.

Ethics statement

This work was carried out in accordance with The Code of Ethics of the World Medical Association (Declaration of Helsinki) for experiments involving humans. All procedures were performed in compliance with relevant laws and institutional guidelines and were approved by the appropriate institutional committee(s).

Declaration of Generative AI and AI-assisted technologies in the writing process

Not applicable.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships, which may be considered as potential competing interests: Shelly Senders reports being a vaccine consultant for Pfizer. Piotr Korbal has no competing interests to report. Mary Kline, Noor Tamimi, Allison Thompson, Jelena Drozd, Mark W. Cutler, Peter C. Giardina, James Trammel, Lanyu Lei, Yahong Peng, Wendy Watson, and Kathleen McElwee, are all employees of Pfizer (or were employees at the time of the study) and may hold Pfizer stock or stock options.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.127916>.

Data sharing statement

Upon request, and subject to review, Pfizer will provide the data that support the findings of this study. Subject to certain criteria, conditions, and exceptions, Pfizer may also provide access to the related individual de-identified participant data. See <https://www.pfizer.com/science/clinical-trials/trial-data-and-results> for more information

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